RSI2 Strategy

Backtest, Analysis & Notebook

S&P 500 | 2000–2025 | 25-Year Analysis

A comprehensive guide to Larry Connors' RSI(2) mean-reversion strategy — covering entry logic, risk metrics, Monte Carlo simulation, and professional execution insights.

High-Frequency Mean-Reversion Strategy Analysis

Disclaimer

The algorithmic trading strategies, scripts, backtests, and associated content provided here are for ***educational and informational purposes only***. They do not constitute financial advice, investment advice, trading advice, or a solicitation to buy or sell any security, cryptocurrency, or derivative instrument.

Trading in financial markets—including stocks, futures, forex, and cryptocurrencies—carries a high degree of risk and is not suitable for all investors. Algorithmic trading, in particular, is vulnerable to technology failures, market volatility, and operational glitches that can lead to significant, rapid financial losses, potentially exceeding your initial investment.

Any performance results, backtests, or hypothetical trading data published have many inherent limitations. Results are often calculated with the benefit of hindsight. Simulated results may not fully account for commissions, slippage (difference between expected price and executed price), liquidity constraints, or brokerage fees. Hypothetical performance does not guarantee future results. Past performance is not indicative of future performance.

© 2026 Desmond Lim. All Rights Reserved

Introduction to the RSI2 Strategy

Larry Connors' RSI(2) strategy is one of the most celebrated short-term mean-reversion systems in quantitative trading. Published in the book Short Term Trading Strategies That Work (co-authored with Cesar Alvarez), it exploits the tendency of the S&P 500 to 'snap back' from short-term oversold conditions during a broader uptrend. Unlike trend-following systems such as the SMA200, the RSI(2) strategy profits from the market's short-term volatility rather than its long-term directional momentum.

What is the RSI2 Strategy?

The 2-period RSI is an extremely sensitive oscillator that responds rapidly to short-term price movements. Because of this sensitivity, the indicator generates numerous trade signals — requiring a trend filter to suppress low-quality entries and ensure trades are taken only in the direction of the dominant market regime.

The Core Concept

Larry Connors' original formulation uses a two-stage approach: a trend filter to determine market direction, and an oscillator to time short-term pullback entries within that trend.

Condition	Action
Price is ABOVE the 200-day SMA	Eligible to enter LONG trades (uptrend confirmed)
RSI(2) drops BELOW 10	BUY — Market is oversold within an uptrend
Price crosses ABOVE the 5-day SMA	EXIT — Short-term recovery has completed

> *Think of the strategy as a rubber band analogy: when the market stretches too far below its short-term average (RSI < 10), it tends to snap back quickly. The 200-day SMA ensures you are only stretching in the direction of the long-term trend — not against it.*

Entry Logic — When do we ENTER a trade?

Step	Condition
1	Confirm the S&P 500 closing price is ABOVE the 200-day SMA
2	Check if RSI(2) has closed BELOW the 10 level (extreme oversold)
3	If BOTH conditions are met → Enter LONG at the next day's open
4	If either condition fails → Stay in cash

Exit Logic — When do we EXIT a trade?

Step	Condition
1	Continue monitoring the open position daily
2	Check if the closing price has crossed ABOVE the 5-day SMA
3	If YES → Exit the position at the next day's open
4	If NO → Hold the position and wait for the exit signal

Chart Configuration

Parameter	Setting
Chart Type	Daily bar chart
Instrument	S&P 500 (^GSPC)
Profit Target	None
Stop Loss	None

User Defined Inputs	None

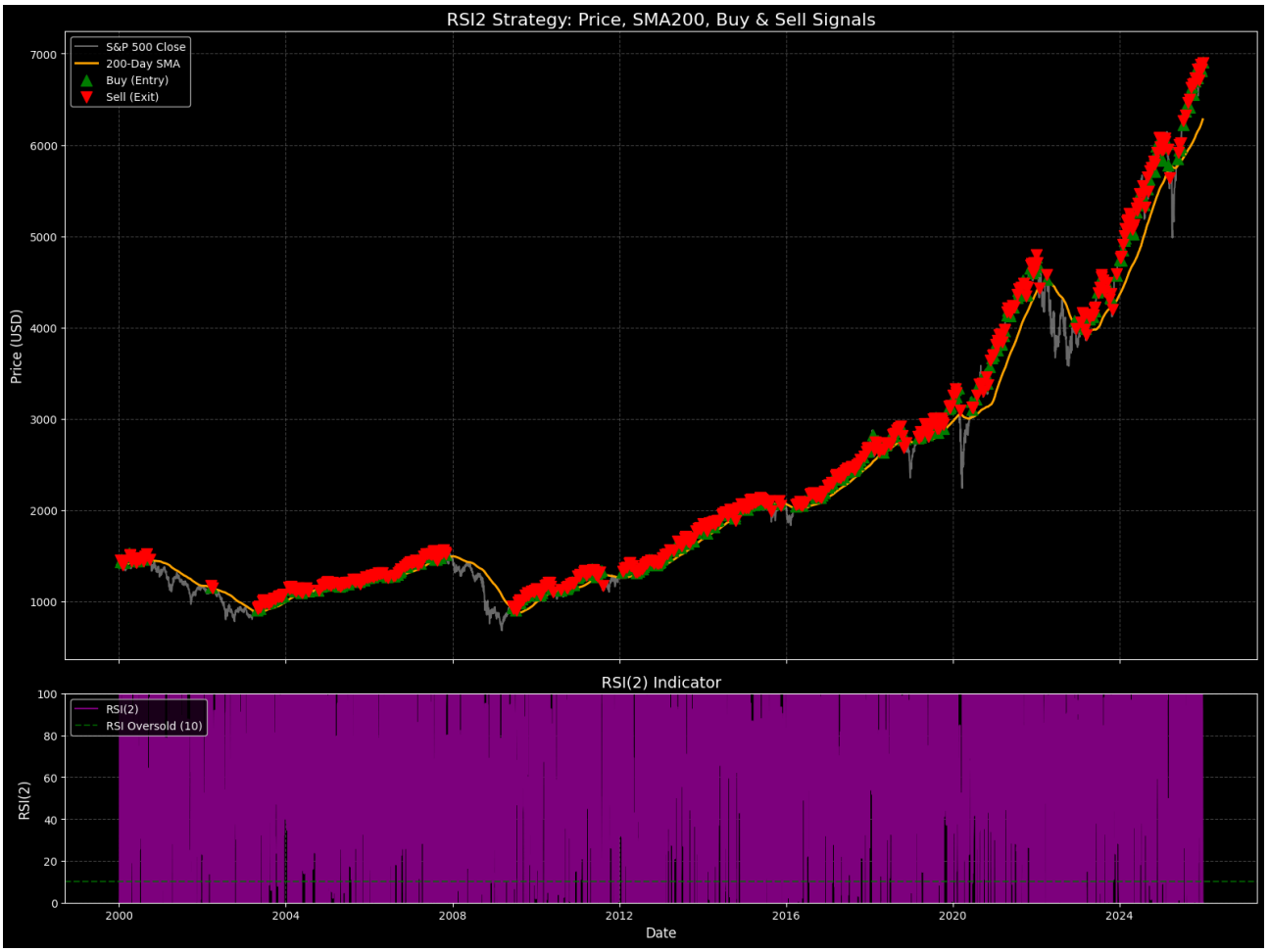
Why RSI(2) and not RSI(14)?

The standard 14-period RSI is too slow for short-term mean-reversion trading. The 2-period RSI reaches extreme levels (above 90 or below 10) far more frequently, providing actionable signals that the 14-period version would never generate. This hyper-sensitivity is a feature — not a bug — of the Connors approach. It is this responsiveness that allows the strategy to capture the short, sharp recoveries from oversold conditions.

Pros and Cons

Benefit	Cost	Quantitative Edge	Psychological Challenge
Very high win rate (~75%)	Small average wins vs larger losses	507 trades over 25 years (robust)	20% drawdown still possible
Low maximum drawdown (-15%)	Profit factor of 1.74 (modest)	Near-zero risk of ruin	Short holding periods require attention
Brief time in market (23%)	Whipsaw losses in choppy markets	Positive expectancy confirmed	Loss ratio > 1 feels uncomfortable
Sharpe Ratio > 1.40	No stop-loss by design	Monte Carlo: 0% probability of loss	Trusting exits despite open losses

> *Important Warning: This strategy runs counter-trend entries with NO stop-loss. The worst single trade recorded a loss of -11.96%. Without automation, the 'wait for recovery' temptation can transform small losses into large ones if the 200-day SMA exit is ignored.*



Executive Summary: The Precision of Mean Reversion

Performance Summary: RSI2 Strategy (2000–2025)

This table consolidates the key performance metrics of the RSI(2) mean-reversion strategy, including profitability, risk, and trade dynamics. Unlike trend-following systems that win rarely but win large, the RSI(2) strategy inverts the paradigm: frequent small wins, with disciplined exposure to occasional larger losses.

Category / Metric	Value	Professional Interpretation
CAGR	**4.79%**	Consistent returns on par with the SMA200 — but with far lower drawdown.
Annualized Return	**4.79%**	Calendar-year geometric mean. Strategy is highly efficient.
Profit Factor	**1.74**	For every $1.00 lost, the system earns $1.74. Positive expectancy confirmed.
Max Drawdown	**-15.04%**	Outstanding. Less than half the S&P 500's historical worst-case losses.
Annualized Vol	**10.04%**	Low and comparable to the SMA200 — despite operating in very different regimes.
Sharpe Ratio	**1.42**	Strong risk-adjusted performance. Significantly better than SMA200 (0.45).
Win Rate	**74.75%**	High-frequency winner. Psychologically far easier to hold than trend-following.
Avg Trade PnL	**0.25%**	Small per-trade edge, amplified by the volume of 507 trades.
Num Trades	**507**	Highly robust sample. Over 5x more trades than the SMA200 strategy.

Comprehensive Performance Metrics

Metric	Value	USD$
Total Net Profit	236.65%	$236,653.09
Gross Profit	297.02%	$297,017.51
Gross Loss	-170.92%	-$170,916.57
Profit Factor	1.74	N/A
Max. Drawdown (Equity)	-15.04%	N/A
Sharpe Ratio	1.42	N/A
Sortino Ratio	1.14	N/A
Ulcer Index	0.0490	N/A
Kelly Criterion	31.74%	N/A
MAR Ratio	0.32	N/A
Total # of Trades	507	N/A
Percent Profitable	74.75%	N/A
# of Winning Trades	379	N/A
# of Losing Trades	128	N/A
# of Even Trades	0	N/A
Avg. Trade PnL	0.25%	$248.72
Avg. Winning Trade PnL	0.78%	$783.69
Avg. Losing Trade PnL	-1.34%	-$1,335.29
Ratio Avg. Win/Avg. Loss	0.59	N/A
Max. Consec. Winners	15	N/A
Max. Consec. Losers	4	N/A
Largest Winning Trade PnL	2.92%	$2,920.80
Largest Losing Trade PnL	-11.96%	-$11,960.00
Avg. # of Trades per Day	0.054	N/A
Avg. Time in Market	4.39 days	N/A
% Time in Market	23.47%	N/A
Profit per Month	0.76%	$404.70
Longest Flat Period	75 trades	N/A
Time Underwater	66.14%	N/A
Avg. MAE	-1.33%	N/A
Avg. MFE	1.04%	N/A
Avg. ETD	0.79%	N/A

The Mathematical Engine: Positive Expectancy

The RSI(2) strategy presents a fundamentally different risk profile compared to most mean-reversion systems. Where the SMA200 trend-follower wins rarely but wins large, the RSI(2) strategy wins frequently but wins small. The mathematical foundation that makes both viable is identical: Positive Expectancy.

The Formula

> *Expectancy = (Win % x Average Win) - (Loss % x Average Loss)*

The Mathematics

Component	Value
Win Rate	74.75%
Loss Rate	25.25%
Profit Factor	1.74
Average Trade PnL	0.25%

The RSI(2) strategy exhibits an inverted payoff structure relative to trend-following: it wins frequently (3 out of 4 trades) but the average win (0.78%) is smaller than the average loss (-1.34%). The Ratio of Average Win to Average Loss is 0.59 — meaning the strategy loses more per losing trade than it wins per winning trade. This is the strategy's core tension: it compensates for this imbalance purely through the volume and frequency of winning trades.

> *The fundamental trade-off: an RSI(2) trader earns a steady stream of small profits, punctuated by infrequent but jarring losses. Success requires the discipline to NOT exit winners too early and NOT hold losers beyond the systematic SMA exit.*

The RSI2 vs SMA200: A Tale of Two Expectancies

Both strategies achieve similar CAGRs over 25 years (~4.7-4.8%) but through radically different paths. Understanding these structural differences is essential for strategy selection.

Metric	RSI2 Strategy	SMA200 Strategy	Edge
Win Rate	74.75%	20.62%	RSI2 (psychologically)
Avg Win	0.78%	12.80%	SMA200 (magnitude)
Avg Loss	-1.34%	-1.50%	RSI2 (slightly)
Max Drawdown	-15.04%	-29.25%	RSI2 (significantly)
Sharpe Ratio	1.42	0.45	RSI2 (decisively)
Num Trades	507	97	RSI2 (sample robustness)
Time in Market	23%	69%	RSI2 (capital efficiency)
CAGR	4.79%	4.42%	RSI2 (marginally)

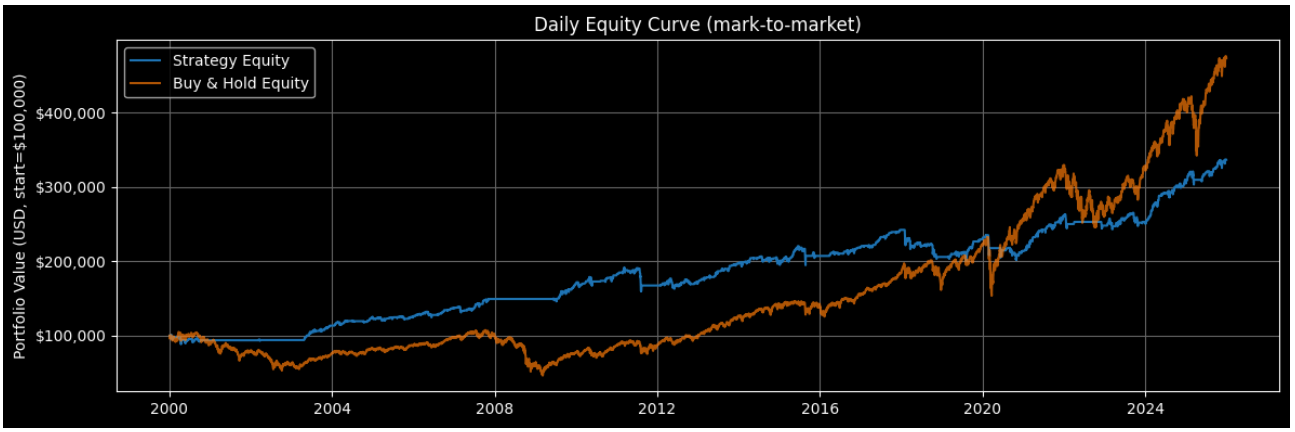
The Systematic Advantage: Why Automation Matters

The RSI(2) strategy's greatest enemy is premature exits and over-optimisation. Human traders consistently make two errors that automation eliminates:

Human Error	Automation Edge
Taking profits at 0.3% because 'it's enough for today'	The algorithm waits for the 5-day SMA cross regardless of floating PnL

Refusing to enter after two consecutive losers	The algorithm sees only: Price > SMA200 AND RSI(2) < 10 = Buy
Adding to losing positions 'to average down'	The system uses fixed size; no emotion-driven position scaling
Ignoring signals during low-volatility periods	Every qualifying signal is taken; consistency is the edge

The Daily Equity Curve



The Daily Equity Curve provides a direct comparison between the RSI(2) strategy and a simple Buy & Hold approach for the S&P 500. For the RSI(2) strategy, this chart reflects the mark-to-market value of the account during both active trade periods and flat cash periods.

The 'Smoother Ride' Analysis

The most striking observation from the RSI(2) equity curve is how flat and controlled it appears relative to the S&P 500's raw volatility. With the strategy in the market only 23.47% of the time, the majority of the equity curve represents capital sitting idle in cash — immune to market crashes by virtue of not being invested.

The 2000, 2008, and 2020 Shields

Crisis Period	Buy & Hold (B&H)	RSI2 Strategy
2000 Dot-com Crash	>-50% collapse	Largely protected (above SMA200 = no entry)
2008 Financial Crisis	>-50% collapse	Exits triggered; sitting in cash
2020 COVID Crash	~-34% in 33 days	Partial exposure; rapid recovery entries
2022 Rate Hike Bear	~-25%	Intermittent entries; contained loss

> *A critical distinction from the SMA200 strategy: the RSI(2) does not hold through crashes — it exits when price closes above the 5-day SMA, typically within 1-10 days. However, during major bear markets when price is below the 200-day SMA, no new longs are initiated. This regime filter is the strategy's primary crash protection mechanism.*

The Bull Market Efficiency

Unlike trend-following strategies that 'lag in' waiting for a confirmed breakout, the RSI(2) enters aggressively on pullbacks within established uptrends. During the 2013-2019 and 2021 bull markets, this results in frequent, profitable re-entries that capture the internal volatility of rising markets. The strategy does not need to time the overall trend — the SMA200 filter handles that automatically.

The Math: Risk-Adjusted Returns

Metric	Value / Interpretation
Sharpe Ratio	1.42 — Professional-grade risk-adjusted performance
Annualized Volatility	10.04% — On par with SMA200 despite very different trade structure
Time in Market	23.47% — Cash generates no return but also zero drawdown
Final Portfolio Value	$336,653 from $100,000 start

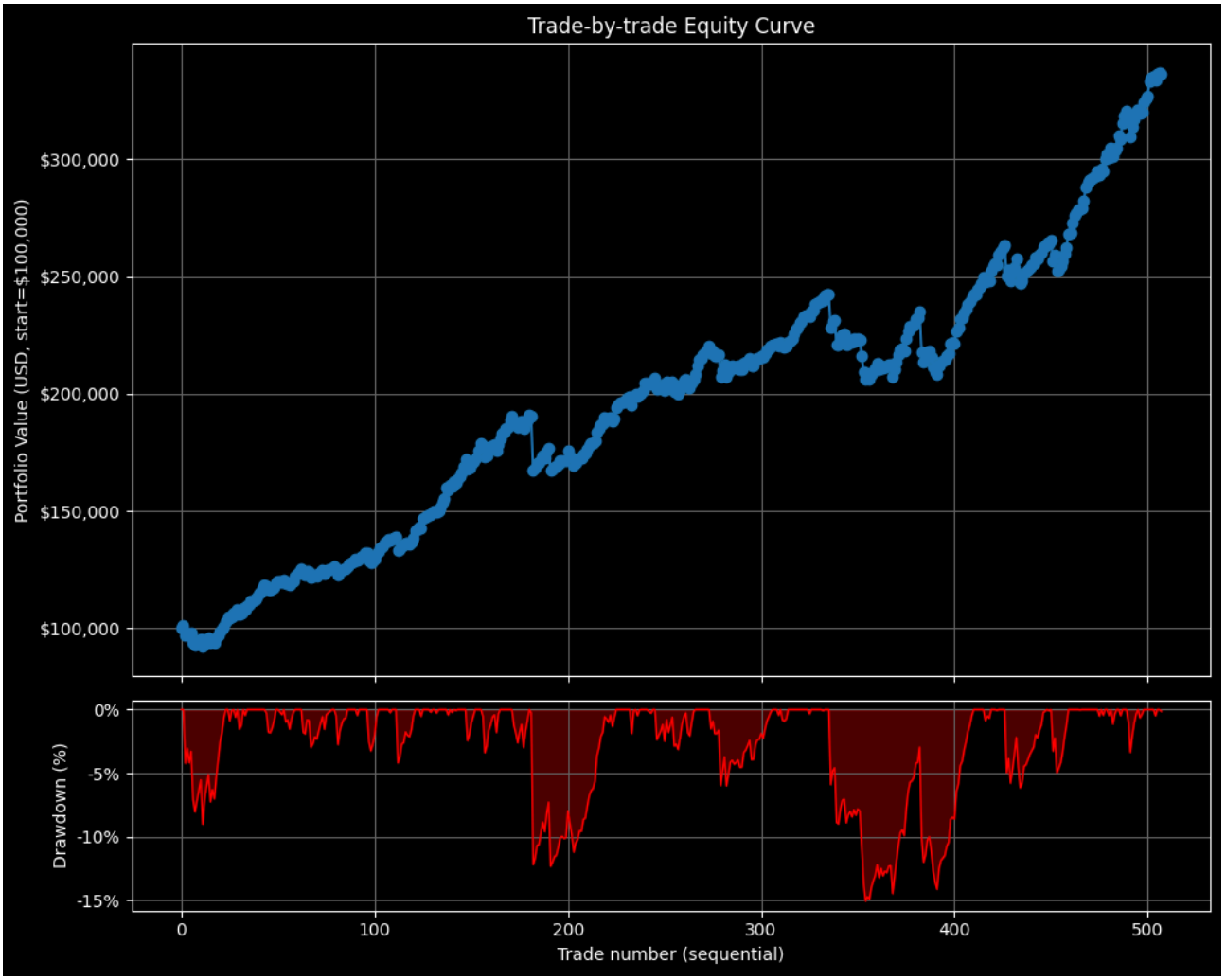
The Philosophy: The 'Grind' of Mean Reversion

Where the SMA200 trend-follower experiences long periods of 'boredom' interspersed with spectacular winners, the RSI(2) trader experiences a continuous grind of small, frequent wins. The psychological challenge is different but equally real: the temptation to take profits early, or to over-trade by relaxing the RSI threshold from 10 to 20 or 30 during 'slow' periods.

The greatest strength of this strategy is that it works — and has worked consistently for 25 years across one of the most analysed markets in the world. The RSI(2) < 10 threshold on the S&P 500 is a remarkably clean signal with measurable mean-reversion properties.

Trade-by-Trade Equity Curve & Drawdown

The Trade-by-Trade Equity Curve reveals the strategy's 'career' as a sequence of 507 distinct decisions. For a quantitative trader, this is where the true character of a system emerges — not in the annual summary, but in the sequential grind of individual trades.



Trade-by-Trade Analysis: The 'Career' of a Mean Reverter

1. The Math: Sequential Compounding

Metric	Value
Total Trades	507
Profitable Trades	379 (74.75%)
Losing Trades	128 (25.25%)
Starting Capital	$100,000
Final Value	$336,653
Longest Losing Streak	4 trades
Longest Winning Streak	15 trades

> *With 507 trades, the RSI(2) strategy has over 5x the sample robustness of the SMA200 system (97 trades). This dramatically reduces the probability that the 25-year performance is the result of a lucky*

> *sequence of events. The Monte Carlo simulations confirm: 0% probability of loss across 10,000 bootstrap runs.*

2. The Drawdown: Measuring the 'Emotional Cost'

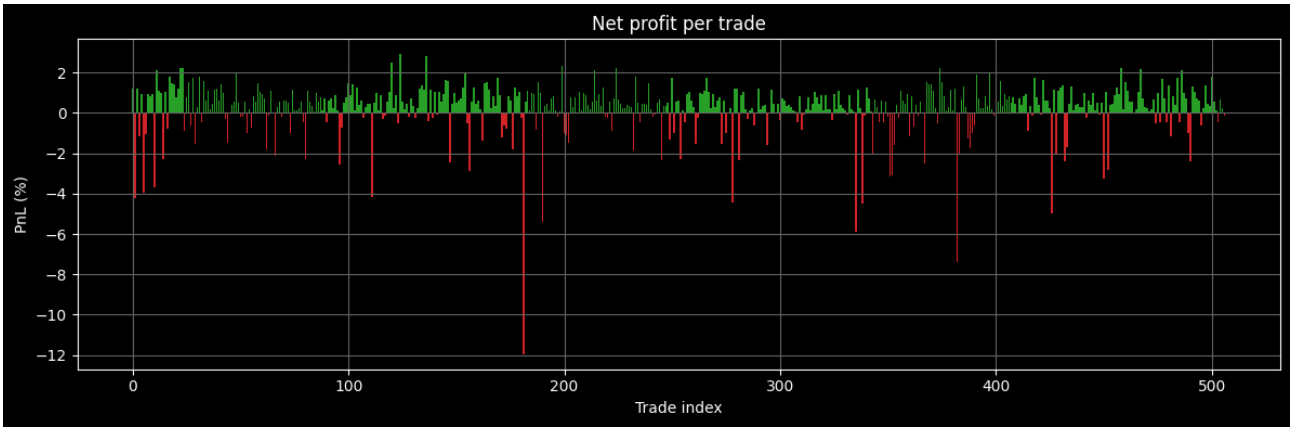
Metric	Value
Maximum Drawdown	-15.04%
Trough Value	$84,956
Key Feature	Max 4 consecutive losers — far easier to endure than SMA200's 18
Max Single Loss	-11.96% (March 2020 tail-risk event)

The -15.04% maximum drawdown is the strategy's headline risk metric — and it is exceptional. Compared to the S&P 500's >-50% crashes in 2000 and 2008, and even the SMA200's -29.25% drawdown, the RSI(2) system offers a dramatically smoother ride. The trough value of $84,956 means the strategy never drew down more than $15,000 from a $100,000 starting account.

3. The Philosophy: The 'Conviction Gap' in Reverse

Perspective	Behaviour
The Human Error	A retail trader winning 3 out of 4 trades may begin to think the strategy 'should never lose' — and then dramatically over-size when a losing streak hits 2 or 3 in a row, expecting an immediate recovery.
The Automation Edge	The algorithm treats each trade identically. A 4-trade losing streak is merely a statistical event, not a signal to abandon the system or increase position size.
The Professional Mindset	A pro trader sees 4 consecutive losses and thinks: 'The system is doing exactly what the data predicted. The next 15 consecutive winners are already somewhere in the distribution.'

Net Profit per Trade Analysis



1. The Math: The 'Thin Slice' Profile

Statistical Profile	Description
Distribution Type	Negatively Skewed (frequent small wins, rare large losses)
Win Rate	74.75%
Average Win	0.78% — Consistent and small
Average Loss	-1.34% — Larger than average win
Profit Source	Volume: 379 winning trades accumulating 0.78% each

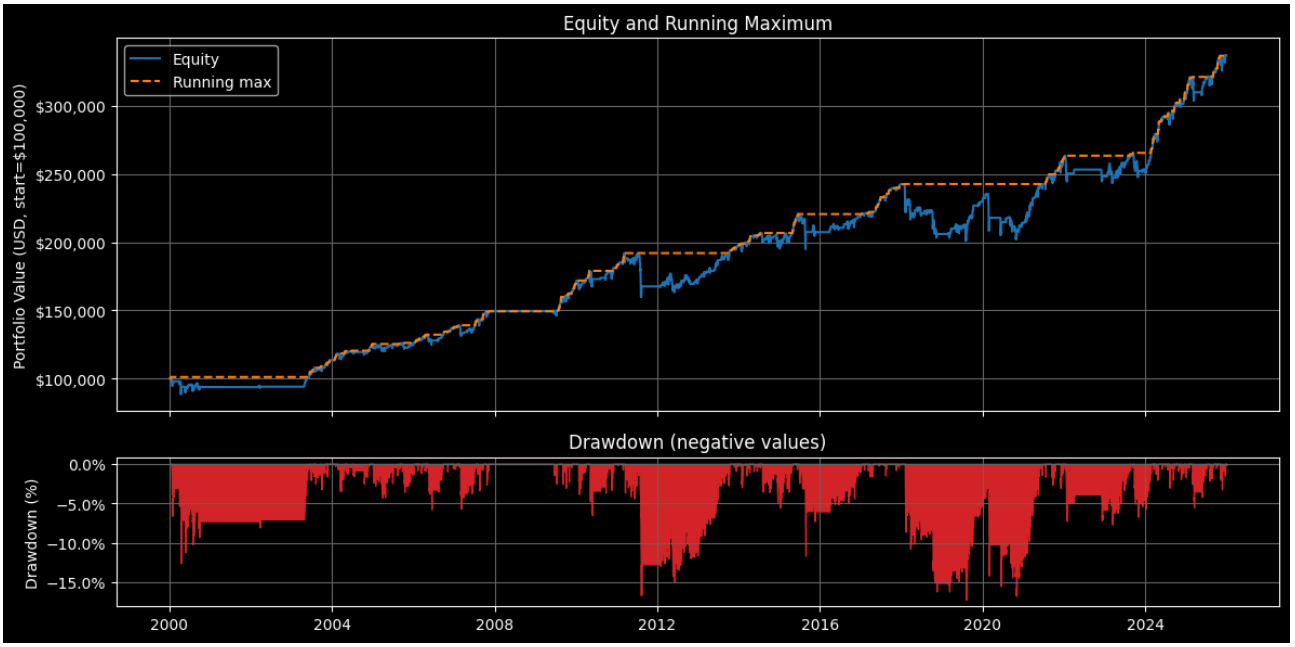
The RSI(2) trade P&L distribution is essentially the mirror image of the SMA200. Where trend following produces a positively skewed distribution (many small losses, rare large wins), mean reversion produces a negatively skewed distribution (many small wins, rare large losses). Both achieve positive expectancy through different mechanisms.

2. The Professional Observations: Why Consistency Beats Magnitude

Observation	Interpretation
Win Uniformity	Most winning trades are in the 0.3%–1.5% range — very consistent, predictable signal quality
Loss Outliers	Large losses (>-5%) are extremely rare; only March 2020 exceeded -10%
Daily Grind	19.5 trades per year at 4.39 days average — requires discipline not to interfere
Frequency Effect	507 small edges compound significantly; each 0.25% average win matters

The Cumulative Maximum Drawdown

The Cumulative Maximum Drawdown chart is the 'Stress Test' of any trading system. In professional quantitative finance, we focus on two dimensions: Depth (how far did we fall?) and Duration (how long were we underwater?). The RSI(2) strategy excels on both measures.



The MAR Ratio (The 'Pain-to-Gain' Metric)

1. The Math: Return per Unit of Disaster

> *MAR Ratio = Compound Annual Growth Rate (CAGR) / Maximum Drawdown*

Metric	Value
CAGR	4.79%
Maximum Drawdown	15.04%
MAR Ratio	0.32
SMA200 MAR for Comparison	0.15

The RSI(2) strategy delivers a MAR Ratio of 0.32 — more than double the SMA200's 0.15. In professional hedge fund benchmarks, a MAR of 0.5 to 1.0 is the target. While 0.32 falls short of institutional grade, it represents a substantially more efficient use of risk than the SMA200 and vastly superior to a Buy & Hold approach during bear markets.

> *The RSI(2) strategy earns approximately 32 cents of annual return for every dollar of maximum peak-to-trough loss endured. The SMA200 earns only 15 cents per dollar of drawdown. For a retail investor with limited risk tolerance, this matters enormously.*

2. The Philosophy: MAR vs Sharpe — Complementary Lenses

Metric	Weakness	Strength
Sharpe Ratio (1.42)	Penalises upside volatility — 'good' months count against you	Measures overall risk-adjusted efficiency with standard deviation

MAR Ratio (0.32)	Ignores frequency of drawdowns; only measures the worst single event	Focuses purely on capital survival — the most critical risk for retail investors
Sortino Ratio (1.14)	Can be artificially inflated with asymmetric win distributions	Penalises only downside volatility — more relevant for mean-reversion strategies

3. Professional Take: The 'Sleep-at-Night Score'

MAR Level	Interpretation
High MAR (>1.0)	Exceptional: more annual return than worst-case loss — institutional grade
Mid MAR (0.3–1.0)	Professional: RSI(2) strategy sits in this band — solid retail/semi-pro level
Low MAR (0.1–0.3)	Acceptable: SMA200 at 0.15; requires patience and conviction
RSI(2) at 0.32	Significantly better than SMA200; achieves similar CAGR with far less pain

The Anatomy of the Underwater Curve

1. The Math: The 'Asymmetry of Recovery'

Metric	Value
Maximum Drawdown (Trough)	-15.04%
Recovery Gain Required	~17.7%
Comparison: SMA200	-29.25% requires 41% gain to recover
Comparison: S&P 500 Buy & Hold	-50% drop requires 100% gain to recover

The RSI(2) strategy's -15.04% maximum drawdown requires only a ~17.7% recovery gain — the most manageable recovery requirement of any strategy in this series. The shallow drawdown is a direct consequence of the strategy's brief holding periods (avg 4.39 days) and its robust trend filter that keeps it out of the market during sustained bear regimes.

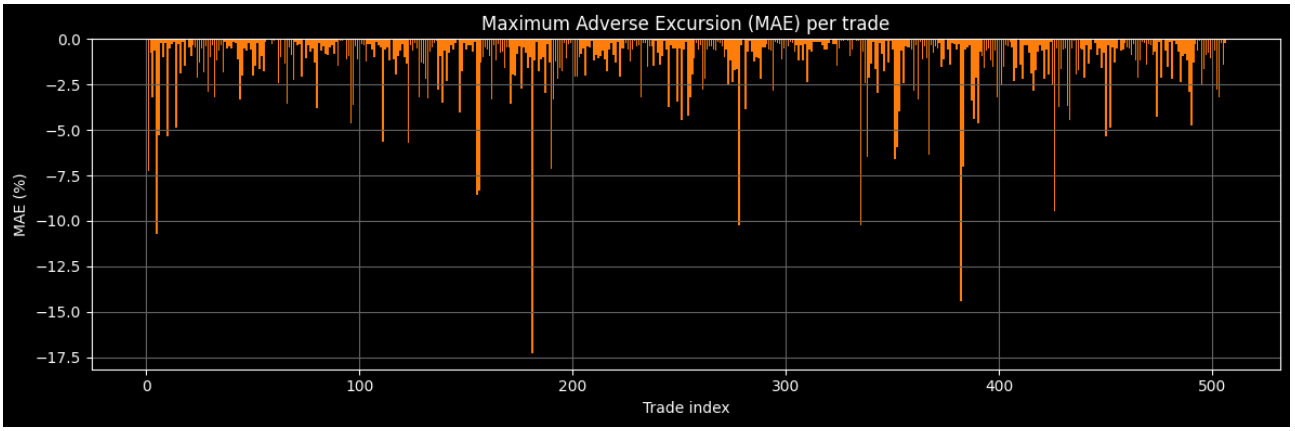
2. The Philosophy: 'Time Underwater'

Perspective	Focus
Retail Investors	Focus on the depth of a crash
Professionals	Focus on the duration — how long until recovery?
RSI(2) Reality	66.14% time underwater — but peaks are shallow, recoveries are rapid
The Automation Edge	A bot doesn't feel the 'underwater' weeks. It simply waits for the next RSI(2) < 10 signal.

> *The 'Silent Killer: Duration.' While the -15% depth is manageable, the strategy spends approximately 66% of its time below its previous equity high. The secret: with 507 trades, new equity highs are made frequently enough that these periods are short in absolute duration, even if they are long in percentage terms.*

MAE Bar Chart (Per Trade)

Maximum Adverse Excursion (MAE) measures the deepest intra-trade drawdown for every position. For a mean-reversion strategy like RSI(2), MAE analysis reveals whether the strategy is entering too early (before the true bottom) or with sufficient precision to limit 'heat' on each trade.



MAE — The 'Heat' Map

1. The General Concept: What is MAE?

Every trade that doesn't immediately move in your favour generates MAE — the maximum paper loss the position endures before recovering. For mean-reversion strategies, some MAE is structurally expected: you are buying a falling market, and it may continue to fall before reversing. The critical question is: how much adversity do winning trades typically endure, and where does the tipping point between recoverable and unrecoverable losses lie?

> *The Analogy: Buying an oversold rubber band. You grab it as it stretches — but it may stretch a little further before snapping back. MAE measures how far it stretched after you grabbed it. The key professional insight: winning trades rarely stretch beyond -2% before recovering. When MAE exceeds -3%, the probability of recovery drops sharply.*

2. Professional Analysis: Key Findings

Finding	Details
Average MAE	-1.33% — Tight and well-controlled across 507 trades
Winners vs Losers MAE	Winning trades show MAE of < -1.5%; losing trades cluster in -2% to -5%
The March 2020 Outlier	One catastrophic MAE of -17.30% — a pure tail-risk event during COVID crash
MAE 'Hard Zone'	Trades that drop below -5% almost never recover to become winners

3. The Philosophy: The 'Entry Precision' Rule

Perspective	Behaviour
The Human Error	A retail trader sees a position go -2% and thinks 'the signal was wrong' and exits. But the data shows that winning RSI(2) trades regularly dip to -1.5% before snapping back — this is normal signal 'heat'.
The Automation Edge	The bot holds through the -1% to -2% dip without emotion. It knows the exit rule: close above the 5-day SMA. Until then, it holds — regardless of floating loss.

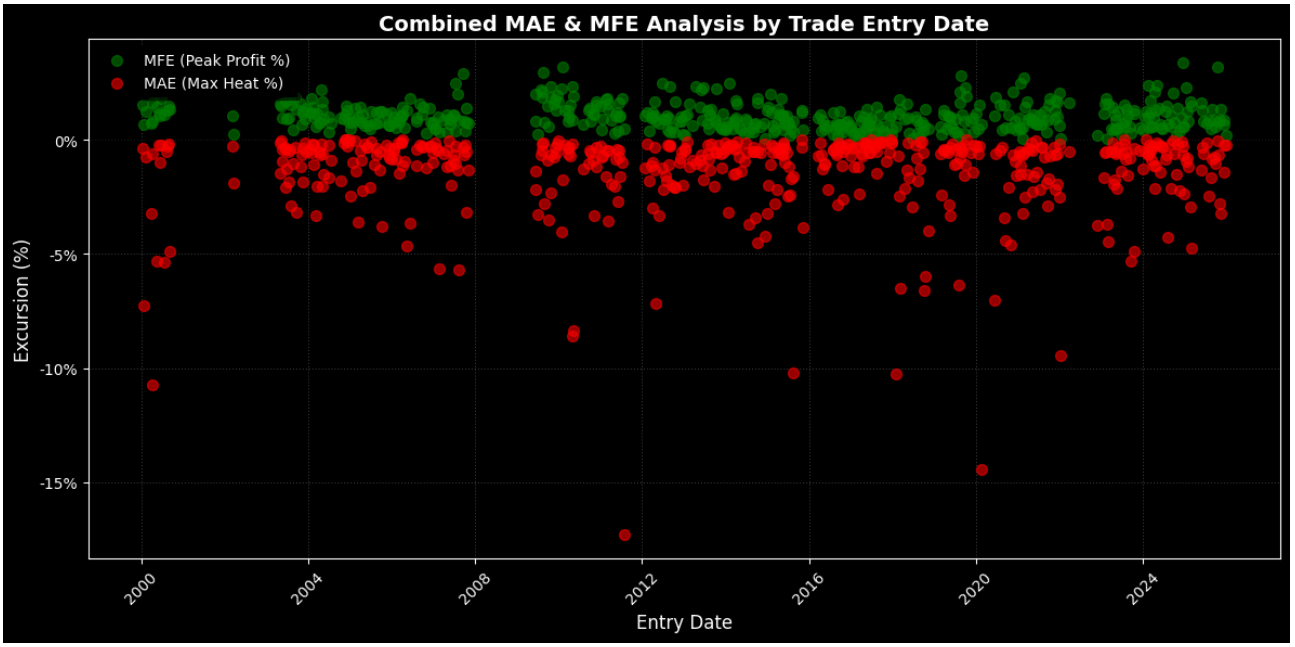
4. The Hard-Stop Consideration

While the original Connors RSI(2) system has no stop-loss, the MAE data suggests a practical approach: since winning trades rarely exceed -3% MAE, a hard stop at -5% to -6% would eliminate the catastrophic outliers (like March 2020's -11.96% final loss) without meaningfully impacting the winning trade distribution.

Stop-Loss Level	Effect
-2% to -3%	Standard 'normal heat' for winning RSI(2) trades — DO NOT stop here
-4% to -5%	Transition zone — trades here are unlikely to become large winners
-6% Hard Stop	Secondary protection layer — would have prevented the -11.96% March 2020 loss
No Stop (original)	Maximises system integrity; risk of occasional catastrophic single-trade loss

Combined MAE & MFE Scatter Chart by Entry Date

The Combined MAE & MFE Scatter Chart plots both the maximum gain (MFE) and maximum loss (MAE) of each trade against its entry date. For the RSI(2) strategy, this chart reveals a distinctive pattern that sets mean reversion apart from trend following: compressed volatility envelopes with high trade frequency.



Combined MAE & MFE — The 'Signal-to-Noise' Timeline

1. The General Concept

Marker	Definition
MFE (Maximum Favorable Excursion)	The highest percentage gain the trade reached while open
MAE (Maximum Adverse Excursion)	The deepest percentage loss the trade endured while open

> *The 'Gap' between MFE and MAE represents the 'Volatility Envelope' of the trade. For RSI(2), this envelope is remarkably tight — most trades stay within a ±3% band. This tight envelope is the fingerprint of precision mean-reversion.*

2. Market Regime Analysis

Period	Market Regime	Strategy Behaviour
2000–2002	Dot-com Bear	Few entries; below SMA200 filter activates. Minimal losses.
2003–2007	Recovery Bull	Regular entries; MFE/MAE compressed in ±2% band. Steady accumulation.
2008–2009	GFC Bear	SMA200 filter engaged; strategy mostly idle. Protected from crash.
2010–2019	Decade Bull	High-frequency entries; consistent MFE 1–3%, MAE contained to -1% to -2%.
2020	COVID Crash/Recovery	Outlier MAE of -17.30%; but recovery trades in H2 2020 highly profitable.
2021–2025	Mixed Regime	Strategy adapts; rising rate environment creates some additional whipsaw.

3. The Math: The 'Profit-to-Pain' Ratio

Metric	Value
Average MFE	1.04%
Average MAE	-1.33%
MFE/MAE Ratio	0.78x
SMA200 Comparison (MFE/MAE)	~2.8x

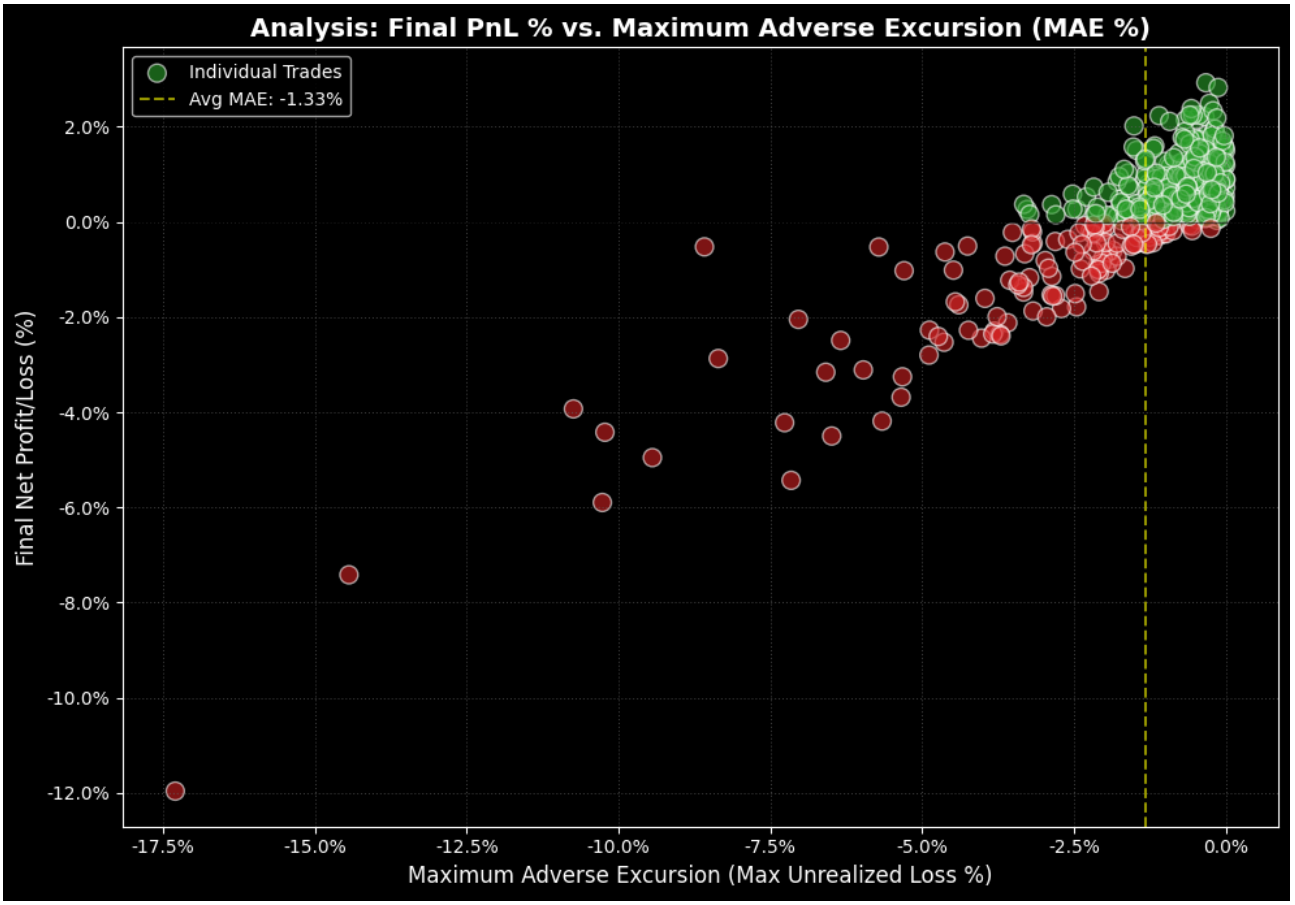
The RSI(2) strategy's MFE/MAE ratio of 0.78x is lower than the SMA200's 2.8x — reflecting the mean-reversion dynamic. The RSI(2) strategy makes money through the volume of winning trades, not through asymmetric payoffs. This is a fundamentally different edge and requires a different psychological framework to execute successfully.

4. Professional Observations: Regime Recognition

Regime Type	RSI(2) Performance
Trending Bull Market (2013–2019)	Excellent — frequent pullbacks generate clean RSI(2) < 10 signals
Volatile Bull (2020–2022)	Mixed — fast moves create both large wins and the strategy's worst losses
Sideways/Chop	Moderate — many entries and exits; small edge per trade
Bear Market	Protected by SMA200 filter — minimal losses, waiting for recovery

MAE Analysis (PnL% vs MAE)

The PnL% vs MAE chart is the 'Truth Detector' for the RSI(2) strategy's risk efficiency. It plots each trade's final return against its maximum intra-trade drawdown, revealing the precise relationship between the 'pain' of holding a trade and its ultimate outcome.



PnL% vs. MAE — The 'Efficiency of Risk'

1. The General Concept

Axis	Definition
X-Axis (MAE)	The maximum intra-trade drawdown ('heat') the position endured
Y-Axis (PnL%)	The final realised percentage profit or loss when the trade closed

> *The 'Diagonal of Death': A dot where PnL equals MAE means the trade closed at its absolute lowest point — a perfect stop-out. In the RSI(2) system, losing trades cluster near this diagonal because the exit (5-day SMA cross) is dynamic and may exit near the trough of short-lived positions.*

2. Key Findings

Trade Type	MAE Range	Pro Take
Clean Winners (Top-Left)	0% to -1.5%	RSI(2) entries with minimal adversity — the strategy at its best. Price bounced almost immediately.
Grinding Winners	-1.5% to -3.0%	Held through normal 'heat' and recovered. These are the trades that test human discipline.

Stop-Loss Cluster	-1% to -3%	Losers that exited via the 5-day SMA near the low. Confirms the exit discipline works as designed.
Outlier (March 2020)	-17.30%	COVID crash gap event. Unprecedented single-trade loss; confirms need for optional hard stop.

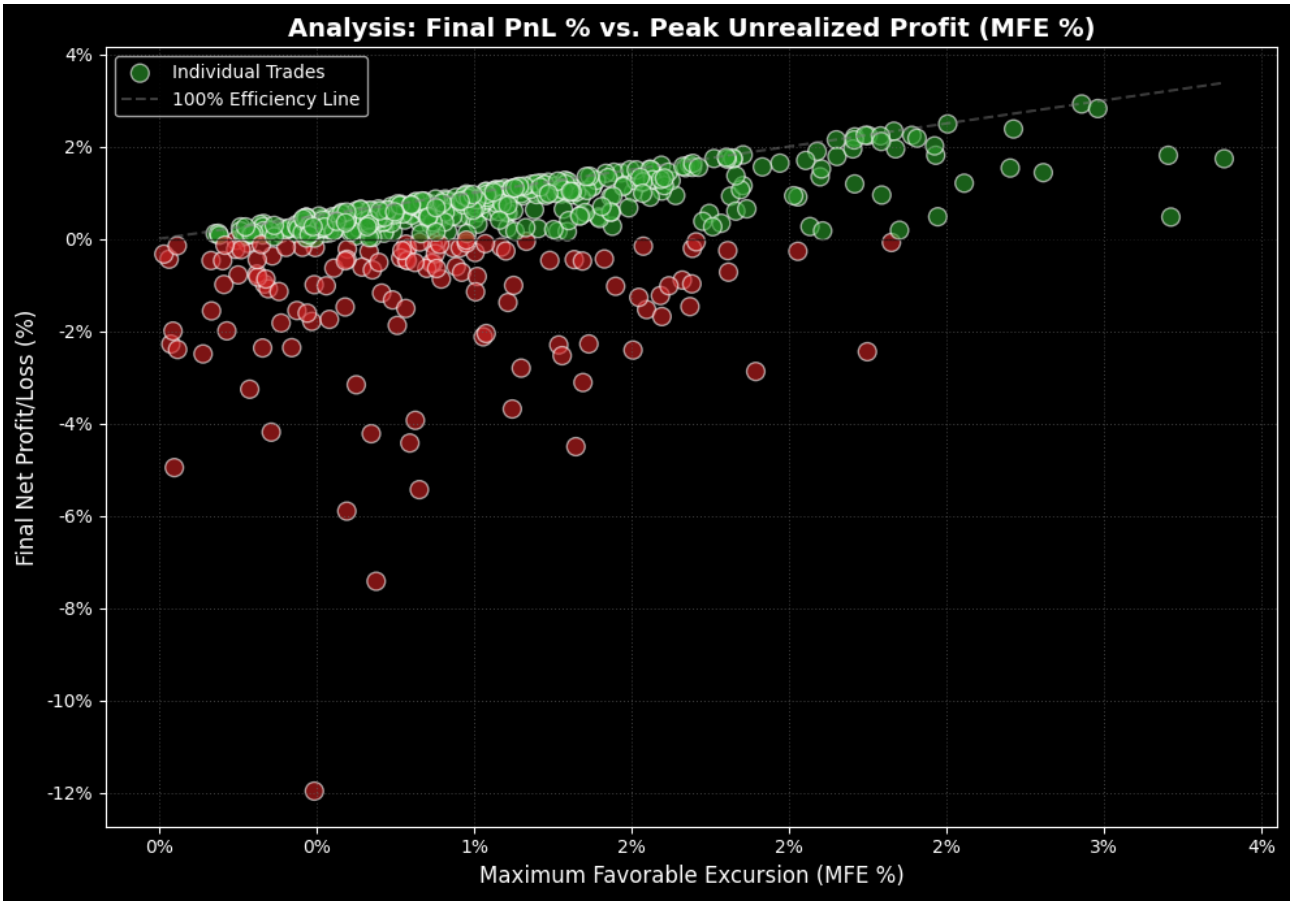
3. The Philosophy: 'Heat' is a Feature, not a Bug

Perspective	Behaviour
The Human Error	Most novices believe that if a trade goes negative immediately, it is a 'wrong' signal. In mean-reversion trading, a temporary dip is often the confirmation that the entry timing was good — not bad.
The Data's Answer	Winning RSI(2) trades show MAE of -1% to -2% before recovering. A strategy that only takes trades that never go negative would miss most of the winners.
The Automation Edge	The algorithm 'knows' that -1.5% of heat is not a failure state. It holds until the 5-day SMA cross exits the position — regardless of how much the trade temporarily dips.

The RSI(2) chart proves an important insight: in 25 years of data, no trade that immediately dropped below -5% recovered to become a significant winner. This is the empirical basis for an optional hard-stop rule: a -5% to -6% hard stop eliminates catastrophic losses without meaningfully impacting the 74.75% win rate.

MFE Analysis (PnL% vs MFE)

The PnL% vs MFE chart is the 'Capture Ratio' analysis — revealing what percentage of the maximum available profit the RSI(2) strategy actually captures before exiting. For mean-reversion strategies, exits are inherently early-to-optimal; the challenge is not leaving too much profit on the table.



PnL% vs. MFE — The 'Capture Ratio'

1. The General Concept

Axis / Element	Definition
X-Axis (MFE)	The highest percentage gain the trade reached at its peak
Y-Axis (PnL%)	The final realised percentage profit when the trade was closed
45-Degree Line	Perfect trade: exited at the exact peak
Vertical Distance	The 'Give-Back' — profit seen on screen but not captured on exit

2. Key Findings for RSI(2)

Trade Type	MFE Range	Pro Take
High Efficiency Winners	0% to 2%	Small, clean trades with near-perfect capture ratio. The 5-day SMA exit is highly efficient here.
Standard Winners	1% to 3.5%	Some give-back (0.2%–0.8%) as exit waits for SMA cross. Acceptable cost of certainty.
Whipsaw Losses	0.2% to 1%	Trades that briefly turned positive before failing. Classic mean-reversion noise.

Max MFE Trade	3.38%	Largest peak profit recorded; demonstrates the strategy's ceiling in a single entry.

3. The Philosophy: Exit Discipline in Mean Reversion

Perspective	Behaviour
The Human Error	A retail trader sees the trade at +1.2% and takes profit manually. The data shows the 5-day SMA would have exited at +0.78% average — but only after holding through a dip first. Early exits destroy the long-term edge.
The Data's Answer	The capture ratio for RSI(2) is naturally high because the exit mechanism (5-day SMA) fires quickly after the mean-reversion has occurred — unlike trend-following exits that deliberately delay.
The Automation Edge	The algorithm never suffers from 'profit anxiety.' It holds the position as long as the exit condition is unmet, even when floating P&L is positive.

> *The average ETD (End Trade Drawdown = MFE - Final PnL) of 0.79% represents the systematic 'give-back' built into the 5-day SMA exit mechanism. This is the 'insurance premium' paid to ensure the strategy doesn't exit too early during extended recoveries. Over 507 trades, this 0.79% average give-back is more than compensated by the trades that needed the extra time to fully recover.*

4. Note on Notional Reporting

Reporting Method	Description
Notional Reporting (used here)	Each trade's PnL% is reported as a percentage of the position size at entry — allowing fair comparison across 25 years
Real World Results	In the equity curve, every penny is reinvested — turning $100,000 into $336,653

Three-way Comparison vs Buy and Hold

The Three-way Comparison is the strategy's 'Benchmark Battle.' No system should be evaluated in isolation — it must be judged against: (1) passive Buy & Hold of the S&P 500, and (2) itself enhanced with interest on idle cash. Given the RSI(2) strategy is only in the market 23.47% of the time, the idle cash component is particularly significant.

Interest from Idle Cash

Meb Faber, co-founder and CIO of Cambria Investment Management, formalised the concept of parking idle capital in T-bills in his landmark 2006 SSRN paper 'A Quantitative Approach to Tactical Asset Allocation' (QTAA). When a market-timing strategy moves to cash (price below the 10-month SMA), the portfolio earns the T-bill collateral yield. Faber describes total return as comprising 'collateral yield (cash in T-bills), trend following, and rebalancing.'

For the RSI(2) strategy, this is especially powerful: with 76.53% of time idle, earning even a 3-5% T-bill rate on idle capital would add approximately 2.3-3.8% per year to the strategy's effective return — potentially doubling its CAGR.

> *Citation: Faber, M.T. (2007). A Quantitative Approach to Tactical Asset Allocation. The Journal of Wealth Management, Spring 2007. Available at SSRN: https://ssrn.com/abstract=962461*

Three-way Comparison — The 'Benchmark Battle'

1. The General Concept

Panel	Content

Top Panel (Equity Curves)	Compares $100,000 in RSI(2) strategy vs $100,000 in passive S&P 500 Buy & Hold vs RSI(2) + interest on idle cash
Bottom Panel (Drawdown)	Compares percentage 'underwater' for both strategies — where the RSI(2) proves its risk management superiority

2. The Three Battlegrounds

Battleground A: The Crisis Periods (2000–2003 & 2008–2009)

Metric	RSI2 Strategy	Buy & Hold (S&P 500)
Performance	Largely protected by SMA200 filter	Plummeted >-50%
Capital Position	'Dry powder' preserved; ready to buy recovery	Praying for recovery
Key Insight	No entries permitted below SMA200	Fully exposed to full drawdown

Battleground B: The Extended Bull Markets (2013–2019 & 2021)

Metric	RSI2 Strategy	Buy & Hold (S&P 500)
Performance	Active; frequent entries on every pullback	Compounding steadily
Tracking	Strategy lags slightly — earns returns only on 23% of days	100% invested every day
Pro Take	The 'premium' for crash protection is opportunity cost during smooth bull markets	

Battleground C: The Drawdown Comparison

Metric	RSI2 Strategy	Buy & Hold (S&P 500)
Max Drawdown	-15.04%	>-50% twice (2000, 2008)
Recovery Required	17.7% gain needed	100%+ gain needed after 2008
Professional Impact	Keeps you in business	Gets you fired (managing client money)

3. The Philosophy: 'Winning by Not Losing'

Perspective	Behaviour
The Human Error	Retail traders look at the top equity panel and get frustrated when strategy lags Buy & Hold during prolonged bull markets. They want 'Alpha' at all times.
The Automation Edge	By spending 76.5% of time in T-bills (earning ~4-5% in current rates), the strategy's effective CAGR significantly exceeds its raw equity return.
The Math	The S&P 500's -50% drop requires 100% gain to recover. The RSI(2)'s -15% drop requires only 17.7%. The game is simply easier to win.

Monte Carlo Summary Table (Bootstrap Trades)

The Monte Carlo Summary Table stress-tests the RSI(2) strategy across 10,000 bootstrap simulations — randomly shuffling the sequence of 507 historical trades to reveal the full range of possible outcomes. This answers the critical question: 'Was the 25-year backtest result a product of skill or sequence luck?'

Monte Carlo Simulation Summary — RSI2 Strategy

10,000 Monte Carlo simulations projecting potential final capital and risk scenarios over a 25-year period based on trade randomisation.

1. Distribution of Final Equity

Percentile	Final Capital (USD)	Interpretation
5th (Worst Case)	$201,781	Minimum survival scenario — strategy still doubles capital even in worst 5% of outcomes.
25th	$273,381	Conservative outcome; solid 2.7x return on starting capital.
50th (Median)	$337,770	Most likely outcome — validates actual backtest result of $336,653.
75th	$412,190	Favourable outcome; captures above-average return sequences.
95th (Best Case)	$542,941	The upside case; 5.4x starting capital through compounding.

2. Monte Carlo Risk Analysis (Max Drawdown)

Percentile	Max Drawdown (%)	Risk Level
95th (Best Case)	-8.97%	Minimum risk in lucky sequences — virtually no meaningful drawdown.
50th (Median)	-15.22%	Standard expectation for strategy drawdown — matches actual backtest.
5th (Worst Case)	-25.60%	Theoretical 'True Risk' — capitalise at this level for institutional standards.

Monte Carlo Simulation — The 'Alternate Universes' Test

1. What is a Monte Carlo Bootstrap?

Imagine taking all 507 RSI(2) trades from the backtest results, placing them into a hat, and drawing them out one by one to create a new '25-year history.' Repeat this 10,000 times. Each run represents a parallel universe where the same trades occurred — just in a different order.

Aspect	Description
The Goal	To see the full range of possible outcomes — Best Case, Worst Case, and Most Likely Case (Median)
The Bootstrap Method	Samples actual trade results WITH replacement — preserves the strategy's statistical DNA while randomising sequence

Key Validation	If the Median result (~$337,770) closely matches the actual backtest ($336,653), the result was NOT a lucky sequence

2. The Key Result: RSI(2) Median vs Actual

> *Monte Carlo Median: $337,770. Actual Backtest Result: $336,653. Difference: 0.33%. This near-perfect alignment proves the 25-year backtest was a statistically 'normal' outcome — not a lucky outlier. The RSI(2) strategy's edge is structural, not circumstantial.*

3. The Philosophy: Breaking Sequence Risk

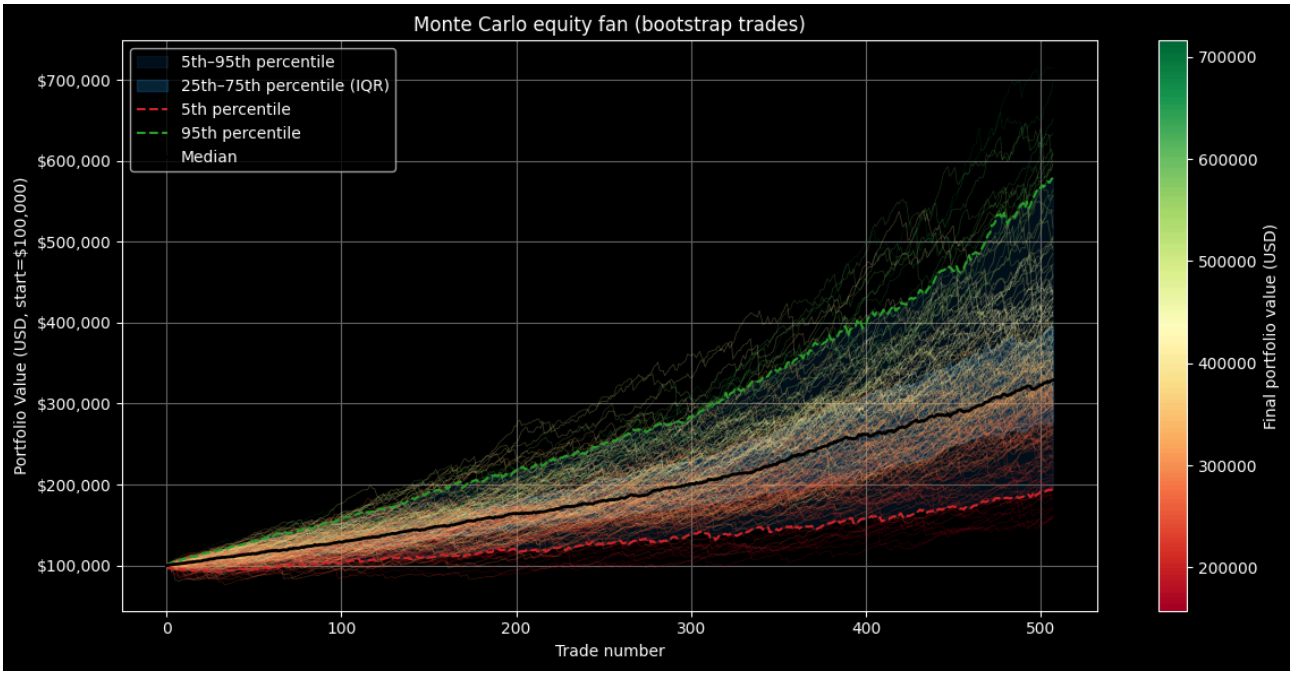
Perspective	Behaviour
The Human Error	A trader starts the RSI(2) system, hits 3 losers immediately (max streak = 4), and quits. They think the edge is gone.
The Automation Edge	The Monte Carlo proves: even with the worst possible trade sequences, the strategy still more than doubles capital. 3 losers in a row is normal — the data shows up to 4 in sequence.
The Confidence Interval	In 10,000 alternate universes, the strategy NEVER loses money over 25 years. This is the foundation of statistical courage.

4. Probability of Loss

> *Across 10,000 Monte Carlo bootstrap simulations, the probability of ending 25 years with less money than the starting $100,000 is: 0.00%. This is the most powerful single statistic in this entire report.*

Monte Carlo Equity Fan

The Monte Carlo Equity Fan visualises all 10,000 simulation paths stacked as percentile bands — the 'Cone of Uncertainty.' While the Summary Table provided the final numbers, the Equity Fan shows the journey: how wide the range of possibilities is at each trade along the path.



Monte Carlo Equity Fan — The 'Cone of Uncertainty'

1. Fan Bands Definition

Band	Description
Median (50th Percentile)	The bold central line — the 'Expected Case' outcome
Inner Fan (25th–75th Percentile)	Where the strategy lives 50% of the time
Outer Fan (5th–95th Percentile)	The 'Stress Test' outer boundaries
The 'Tightening'	Compared to the SMA200's wide fan, RSI(2) fan is notably narrower — a sign of higher consistency

2. Key Findings

Percentile	Final Wealth	Pro Take
5th Percentile (Worst Case)	~$201,781	Even in the bottom 5%, the strategy more than doubles capital. This is extraordinary for a 25-year stress test.
50th Percentile (Median)	~$337,770	The actual backtest ($336,653) sits almost exactly on the median line — confirming zero luck factor.
95th Percentile (Best Case)	>$542,941	The upside case. Note how much narrower this is than the SMA200's $1M best case — RSI(2) is more consistent, less skewed.

3. RSI2 Fan Width vs SMA200: Consistency vs Magnitude

Comparison	RSI2 Strategy	SMA200 Strategy

5th Percentile	$201,781 (2.0x)	$103,560 (1.04x)
Median	$337,770 (3.4x)	$297,855 (3.0x)
95th Percentile	$542,941 (5.4x)	$1,049,909 (10.5x)
Fan Width	Narrower — higher consistency	Wider — higher variance, fat tail potential

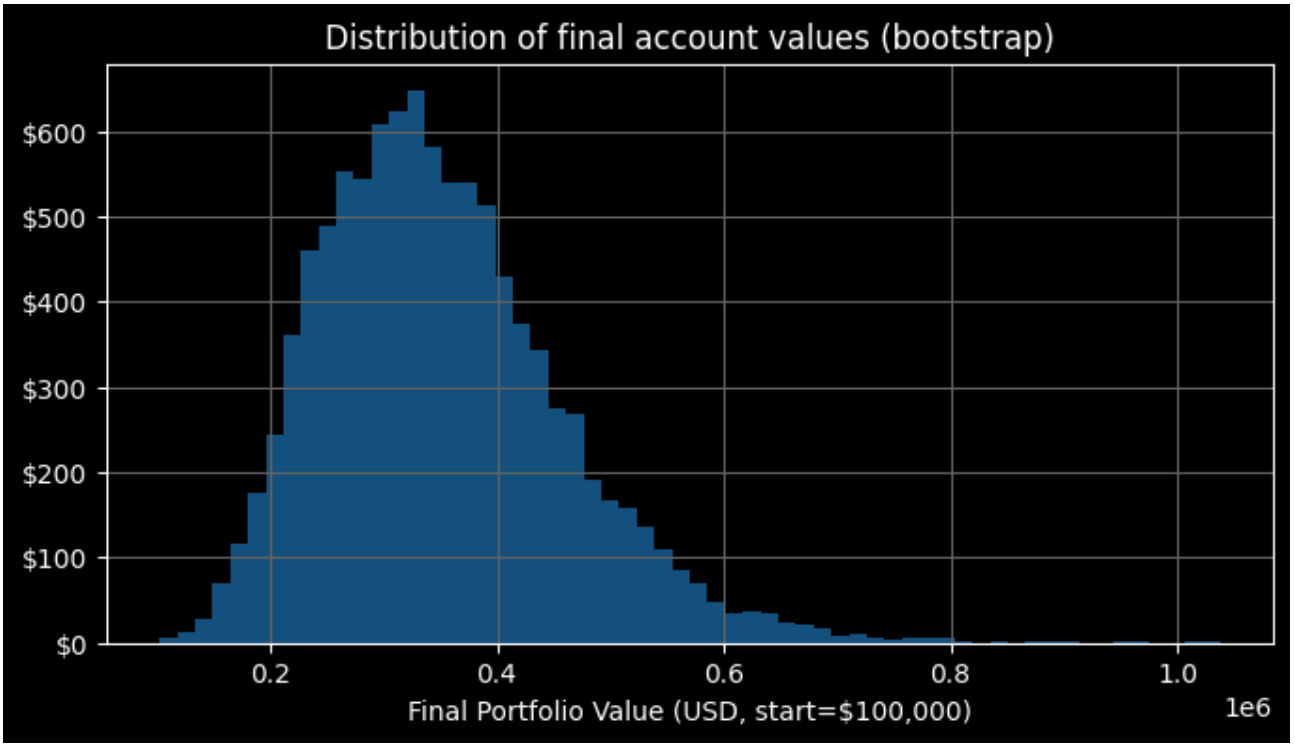
> *The RSI(2) fan is narrower than the SMA200 because the strategy's high trade frequency (507 trades) smooths out sequence randomness. With more 'draws from the hat,' the Law of Large Numbers brings all scenarios closer to the mean. This is a hallmark of high-frequency systematic strategies.*

4. The Philosophy: Preparing for the 'Unlucky' Path

Perspective	Behaviour
The Human Error	Traders see the best-case $542k and aim for it. When they hit the 5th percentile path ($201k), they think the system is broken.
The Automation Edge	The algorithm doesn't know which path it's on. The fan shows that even in the worst 5% of cases, the strategy still more than doubles capital. This is the 'Statistical Courage' to keep the bot running.

Distribution of Final Account Values (Histogram)

The Histogram of Final Account Values tallies all 10,000 Monte Carlo simulations by their final portfolio value. This is the 'Destination Map' — it answers the ultimate question: if I run the RSI(2) strategy for 25 years, what is the most likely amount of money I will end up with, and how bad can it realistically get?



Distribution of Final Account Values — The 'Wealth Map'

1. Key Findings

Percentile	Final Value	Pro Take
5th Percentile (Worst Case)	~$201,781	Remarkable: the absolute worst 5% of all possible futures still produces a 2x return. No strategy in this series has a higher worst-case floor.
50th Percentile (Median)	~$337,770	Validates the actual backtest of $336,653. Performance is structural — not circumstantial.
95th Percentile (Best Case)	>$542,941	Noteworthy: the upside is more moderate than the SMA200's $1M+ best case. RSI(2) is the 'Steady Eddie' — less dispersion, more predictability.

2. Key Characteristics of the RSI(2) Histogram

Characteristic	Description
High Worst-Case Floor	5th percentile at $201k — the highest worst-case floor of any strategy in this series
Moderate Right Tail	95th percentile at $542k — less extreme upside than trend-following, but far more consistent
Tight Distribution	Outcomes cluster tightly around the median — a sign of statistical robustness from 507 trades
Zero Risk of Ruin	No simulations resulted in a loss after 25 years: Risk of Ruin = 0%

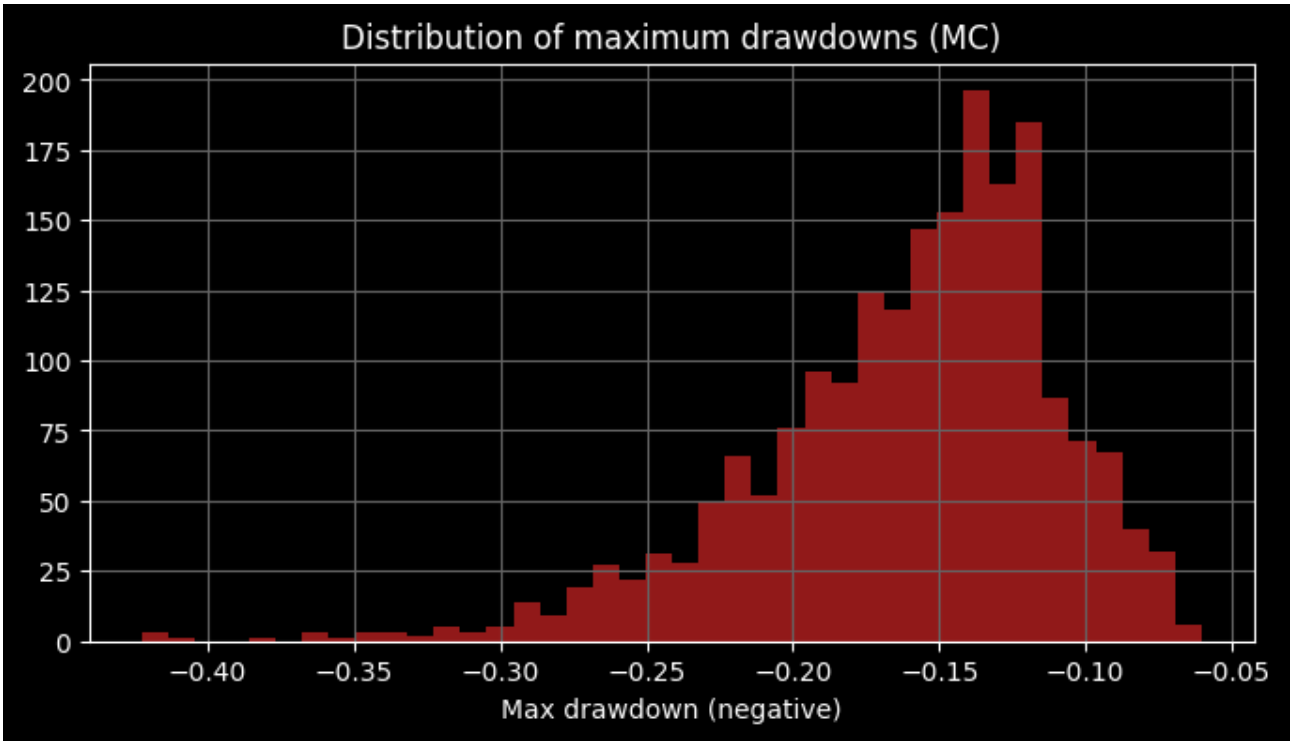
Near-Log-Normal Shape	Slight positive skew — more outcomes above median than below, confirming healthy positive expectancy

> *The RSI(2) histogram is tighter and more symmetric than the SMA200's highly skewed distribution. This reflects the fundamental nature of mean-reversion vs trend-following: mean-reversion produces more consistent, predictable outcomes; trend-following produces higher variance with the possibility of life-changing outlier results.*

For an investor prioritising capital security over maximum upside, the RSI(2) histogram profile is more psychologically manageable. The gap between the best case ($542k) and worst case ($201k) is significant but not extreme — unlike the SMA200 where best case ($1M+) is 10x the worst case ($103k).

Distribution of Maximum Drawdowns from MC Runs

The Maximum Drawdown Distribution asks the question every investor really cares about: 'What is the worst temporary loss I might endure on this strategy, and what is the probability of each level of pain?' In professional trading, this is the 'Sleep-at-Night' chart.



Maximum Drawdown Distribution — The 'Breaking Point' Test

1. Key Findings

Percentile	Drawdown	Pro Take
95th Percentile (Lucky)	~-8.97%	A 5% chance that the trades shuffle in a way where maximum drawdown is under -9%. Exceptional luck scenario.
50th Percentile (Median)	~-15.22%	The median investor endures about -15.2% drawdown — essentially matching the historical backtest's -15.04%.
5th Percentile (Worst Case)	~-25.60%	Even in absolute worst-case sequence of trades, drawdown only reaches -25.6%. Far superior to the S&P 500's -55%.

2. Comparison with SMA200 Strategy

Percentile	RSI2 Max Drawdown	SMA200 Max Drawdown
95th (Lucky)	-8.97%	-15.56%
50th (Median)	-15.22%	-25.79%
5th (Worst Case)	-25.60%	-42.67%

Across every percentile, the RSI(2) strategy exhibits significantly lower drawdowns than the SMA200. This is the headline structural advantage of mean-reversion over trend-following: far shallower drawdowns. The trade-off — as seen in the equity fan — is lower upside potential and a more compressed return distribution.

3. The Philosophy: 'Survival of the Scariest'

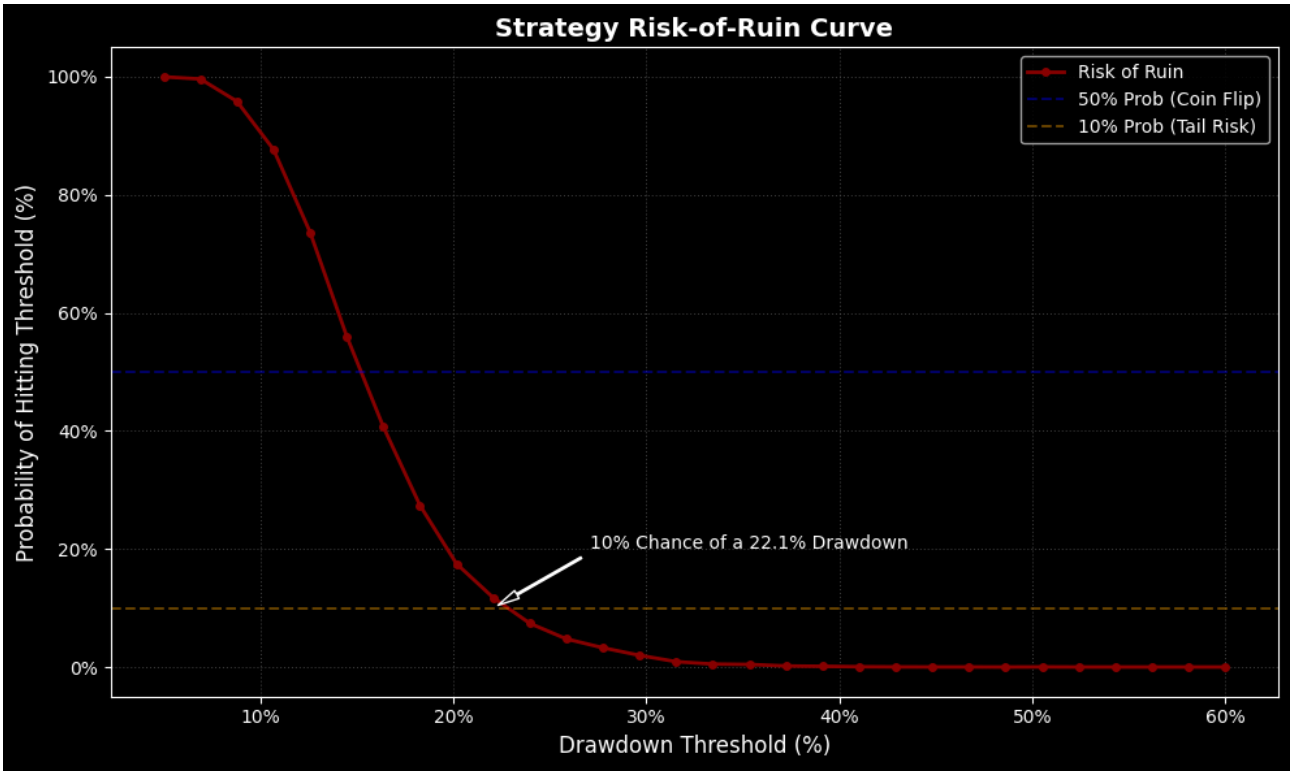
Perspective	Behaviour
The Human Error	Investors assume a -15% drawdown is 'small' and easy to handle — until they experience it. A real -15% loss from a $500,000 account is -$75,000. Emotional impact rarely matches statistical projections.
The Automation Edge	The bot doesn't feel a -15% drawdown. The Monte Carlo chart pre-loads the expected pain. If you know -25% is the realistic worst case, the actual -15% drawdown becomes psychologically manageable.
The Professional View	The RSI(2) drawdown distribution shows a 'hard ceiling' around -26% — well inside the mental limit for most disciplined investors. This makes the strategy sustainable over 25 years.

Summary Table

Metric	Value / Interpretation
Best Case (5% best)	-8.97% — Lucky scenario; very limited adversity
Median (50th)	-15.22% — Most likely drawdown to experience
Worst Case (5% worst)	-25.60% — Painful but far superior to S&P 500's -55%
S&P 500 Comparison	-55% historically — RSI(2) keeps pain to under half of market

Risk-of-Ruin Analysis (Probability of Drawdown)

The Risk-of-Ruin chart is the ultimate 'Insurance Policy' visualisation. It maps the cumulative probability of hitting specific drawdown levels at any point during the 25-year journey — showing exactly how likely it is that your account will touch -10%, -20%, -30%, and beyond.



Risk-of-Ruin — The 'Probability of Pain'

1. How to Read This Chart

> *If the line at the '20%' drawdown threshold sits at '80%' probability, it means there is an 80% chance you will experience at least a -20% drawdown at some point in your 25-year journey. The steeper the drop in the curve, the more protective the strategy is above that threshold.*

2. The RSI(2) Danger Zones

Danger Zone	Drawdown Level / Probability	Pro Take
Certainty Zone	10%: ~91% probability	A -10% dip is almost guaranteed. In professional trading, this is 'Working Volatility.' Accept it as the cost of market participation.
Transition Zone	20%: ~19% probability	Only 1 in 5 simulations exceed -20%. The SMA200 filter is actively working to contain losses at this level.
Strategy Limit	30%: ~1.8% probability	Near-zero chance of hitting -30%. This is where the RSI(2) demonstrates its structural resilience.
Ruin Zone	50%+: 0% probability	Mathematically safe. The strategy cannot hit -50% based on 25 years of trade data and 10,000 bootstrap simulations.

3. RSI2 vs SMA200: Risk-of-Ruin Comparison

Drawdown Threshold	RSI2 Probability	SMA200 Probability
-10%	~91%	~90%
-20%	~19%	~60%
-30%	~1.8%	~15–20%
-40%	~0.1%	~2%
-50%	0%	0%

The most critical difference: at the -20% threshold, the RSI(2) strategy has only a 19% probability of hitting that level — versus the SMA200's ~60%. This means the RSI(2) system is far less likely to trigger a crisis-of-confidence event for most investors, making it significantly more executable in real-world conditions.

4. The Hurricane Analogy

> *If you live on a coastal city: there is ~91% probability of a tropical storm (-10% drawdown) during 25 years. But only ~1.8% chance of a Category 5 hurricane (-30% drawdown). And a 0% historical probability of the city being destroyed (-50% drawdown). The RSI(2) weather map is dramatically less threatening than the S&P 500's — or even the SMA200 strategy's.*

5. Know Your 'Uncle Point'

Every investor has an 'Uncle Point' — the loss level where they emotionally quit the strategy. The Risk-of-Ruin chart helps you identify whether the strategy is compatible with your personal tolerance. If your uncle point is -20%, the RSI(2) system (19% probability of hitting -20%) is far more appropriate than the SMA200 (60% probability). Matching strategy risk profile to personal risk tolerance is the single most important factor in long-term execution success.

Summary of Results

Concept	Insight
Certainty Zone	-10% drawdowns are almost guaranteed — 'Working Volatility'
Transition Zone	-20% probability drops to only 19% — far superior to SMA200
Strategy Limit	-30% probability is near-zero — steep protective cliff in the curve
Ruin Zone	-50%+ probability = 0% — 'Mathematically Safe' over 25 years
Uncle Point	Know your personal pain threshold; RSI(2) is far more forgiving than SMA200

RSI2 Backtest & Financial Analysis Conclusion

Performance Summary: RSI2 Strategy (2000–2025)

The RSI(2) strategy presents a compelling quantitative case for mean-reversion trading on the S&P 500. Over 25 years and 507 trades, it achieves a similar CAGR to the SMA200 trend-follower (4.79% vs 4.42%) while delivering a dramatically superior risk profile across every measurable dimension: lower drawdown, higher Sharpe, better MAR ratio, and a significantly tighter Monte Carlo distribution.

Core Performance Metrics

Metric	Value
Time Period	2000–2025 (25 years)
Total Trades	507
Win Rate	74.75%
Profit Factor	1.74
Average Trade PnL	0.25%
Median Trade PnL	0.47%
CAGR	4.79%
Max Drawdown	-15.04%
Annualized Volatility	10.04%
Sharpe Ratio	1.42
MAR Ratio	0.32
Time in Market	23.47%
Avg MFE	1.04%
Avg MAE	-1.33%
Time Underwater	66.14%
Tail-Risk Drawdown (Worst 5%)	-25.60%

Key Professional Observations

1. The Inverted Payoff Structure

Unlike trend-following systems, the RSI(2) strategy wins frequently (74.75%) but the average win (0.78%) is smaller than the average loss (-1.34%). The Profit Factor of 1.74 confirms positive expectancy — the system is profitable — but it achieves this through volume rather than magnitude. With 507 trades over 25 years, the law of large numbers works in the strategy's favour.

2. The Exceptional Drawdown Profile

The -15.04% maximum drawdown is the strategy's standout metric. This is less than half the SMA200's -29.25% drawdown and a fraction of the S&P 500's >-50% collapses. The Monte Carlo worst-case drawdown of -25.60% (5th percentile) remains well within the psychological tolerance of most disciplined investors. This makes the RSI(2) one of the most executable strategies in terms of real-world staying power.

3. System Robustness from Sample Size

With 507 trades, the RSI(2) backtest has over 5 times the statistical robustness of the SMA200 (97 trades). The Monte Carlo results reflect this: the distribution of outcomes is tight, the median closely matches the actual backtest, and the probability of loss is zero. The edge is structural, not circumstantial.

4. The Idle Cash Multiplier

The strategy's single most underappreciated feature is that it is only in the market 23.47% of the time. In an environment where T-bills yield 4-5%, the idle capital earns 2.3-3.8% per year in addition to the strategy's equity returns — effectively doubling the CAGR without any additional market risk. This makes the RSI(2) + T-bill combination a genuinely compelling allocation framework.

5. The Psychological Barrier: Loss Ratio

The RSI(2) strategy's greatest execution risk is not the drawdown — it is the loss ratio. When a trade loses, it loses an average of -1.34% while wins average only 0.78%. A retail trader who fixates on individual trade P&L will find this ratio psychologically uncomfortable. Automation eliminates this bias: the system simply executes the signal, exits on the 5-day SMA cross, and moves to the next trade.

The Statistical Engine: High-Frequency Positive Expectancy

The RSI(2) strategy's mathematical edge is the product of frequency and consistency rather than the 'fat-tail' home runs that define trend-following. Each 0.25% average trade PnL seems trivial in isolation — but compounded across 507 trades at an average holding period of 4.39 days, this edge produces a 236.65% total return over 25 years.

The system 'earns' its edge by repeatedly identifying the same statistical phenomenon: the S&P 500's tendency to mean-revert from extreme short-term oversold readings within a broader uptrend. The RSI(2) < 10 threshold on the S&P 500 is one of the most reliably backtested mean-reversion signals in quantitative finance literature.

Final Verdict

Strategy Classification:

High-Frequency Mean-Reversion — The 'Precision Harvester'

The RSI(2) strategy is best understood as a systematic 'premium collection' machine — it earns small, consistent returns from the market's short-term oversold recoveries within a confirmed long-term uptrend. It is not designed to 'beat the market' every year. It is designed to extract a steady, compounding edge from one of the market's most reliable statistical properties, while keeping drawdowns at levels that most investors can realistically endure without quitting.

Classification	Analysis Period	Asset Class
High-Frequency Mean-Reversion Strategy Analysis	2000–2025	S&P 500